



What's New in Quantum

Cross-Topic Update Across Five Research Topics

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This report covers what changed across the five quantum topics between each topic's last full edition and 12 September 2026. It is a supplement, not a replacement. The full collections remain the record, and this document reads against them. Each topic has its own window, opening the day after its own edition, so nothing a full edition already published is repeated here as new.

The five topics are post-quantum cryptography migration, quantum computing, post-quantum mathematics, the CA/Browser Forum, and digital asset cryptography. The report opens with the handful of developments that matter most. It then walks the post-quantum migration in four parts: certificates and standards, libraries and silicon, networks and cloud, then sectors and governments. Quantum computing, the mathematics, the CA/Browser Forum and digital assets follow in turn.

Readers of the 3 September migration update will recognise some items in the migration parts. Those are mentioned briefly so the record stays complete. They are not treated as new twice.

1. What Matters Most

Two weeks of updates contain a few developments that change the picture and many that only refresh it, and this section names the ones that change it.

The shape of these two weeks is easy to state. Money that had been promised became money that was signed. One signature candidate fell out of the standards race, and one processor line got a great deal faster. The cloud made post-quantum key exchange the default for a large slice of the web, and governments on three continents said the same thing in the same week. Each of those is a turn in its own story, and the rest of this section takes them one at a time.

Start with the money, where the last editions recorded the CHIPS and Science Act awards to quantum companies as amounts each firm could receive. They are now signed. PsiQuantum finalized an award of \$100 million for optical switches, single-photon detectors and packaging [1]. Rigetti and D-Wave signed definitive agreements, and Quantinuum finalized its award, each worth up to \$100 million [2] [3] [4]. Four companies, one programme, and the word "could" has left the record.

From funding to the algorithms themselves. HAWK, a lattice-based signature candidate, had advanced to the third round of NIST's additional signatures process in May [5]. It was withdrawn on 29 July after a mathematical vulnerability was found [5]. A round-three withdrawal on cryptanalytic grounds is the process working as intended, and it narrows the field that will produce the next standards.



From the mathematics to the machines. IBM's Nighthawk r2 processor adds fast qubit reset and runs more than 100,000 circuits per second [6]. That is 25 times the Heron throughput at Heron-class gate fidelity [6]. It also estimates observables accurately on circuits of more than 7,500 two-qubit gates, up from 5,000 [6]. Throughput is the quantity that decides how much useful work a machine does in a day, and this is a large step in it.

From the machines to the migration that guards against them. AWS Key Management Service now creates and protects ML-DSA keys inside FIPS 140-3 Level 3 validated hardware [7]. The last edition recorded only a CloudHSM preview, so a post-quantum signing key in a managed cloud service has gone from trial to product. At the network edge, Cloudflare's Automatic Key Exchange probes origins and prefers the hybrid post-quantum group [8]. That raised post-quantum origin connections to about 45 billion a day and cut handshake retries from about 52 percent to 3.7 percent [8].

Finally, from the engineers to the governments. The G7's cybersecurity working group called for immediate migration [9]. The US Treasury created a Quantum-Readiness Task Force for the financial sector [10]. India's securities regulator said it is preparing its markets [11]. Three bodies with no shared secretariat reached the same conclusion inside a week, which is the clearest signal in this report that the deadline conversation has moved from technical circles to policy ones.

2. Post-Quantum Migration: Certificates, Identity and Standards

Certificates bind a public key to a name, and everything from web browsing to code signing depends on them.

The certificate debate reported on 3 September has not moved. DigiCert's recommendation of pure ML-DSA certificates over composite ones stands [12]. So does BeatQuantum's timestamp authority signing with pure ML-DSA-87 or SLH-DSA [13]. Microsoft's phased plan for its certificate services is unchanged [14]. DocuSign's ML-DSA rollout still waits on external PDF verifiers before it names a date [15].

The standards drafts kept moving. The composite key encapsulation draft reached version 21 and sits in IESG evaluation with two DISCUSS positions to resolve [16] [17]. The composite ML-DSA draft for TLS authentication reached version 12 [18]. The composite signatures draft for X.509 reached version 19 [19]. HashiCorp Vault Enterprise 2.1.0 added SLH-DSA to hybrid sign and verify operations in its transit engine [20].

The hardware that holds keys moved too. AWS KMS now supports ML-DSA keys in all three parameter sets inside FIPS 140-3 Level 3 HSMs, replacing the CloudHSM preview recorded earlier [7]. Utimaco launched the CSe-Series general purpose HSM with active tamper protection and up to 31 tenant containers, with Level 4 validation in progress [21]. Crypto4A's QASM module completed Level 3 validation [22], and Apricorn's Aegis Secure Key 3.0 did the same [23]. Red Hat's NSS module for RHEL 8 was validated [24].

Smaller tools filled in around the edges. A Rust file-encryption library can re-wrap a file's session key under a new ML-KEM recipient without touching the payload [25]. An npm package keeps ML-DSA and ML-KEM keys non-extractable on PKCS#11 tokens [26]. A .NET package replaces RSA wrapping of ASP.NET data-protection master keys with ML-KEM, aimed at archived data [27].

Quantum key distribution stayed adjacent rather than central. IonQ and Congruity360 signed a deal worth \$8.18 million to deploy QKD pairs alongside post-quantum cryptography [28]. India's C-DAC is building a QKD

sandbox for bank-to-bank transactions [29], and C-DOT unveiled fourteen quantum products including several QKD systems [30]. A preprint demonstrated a three-pair twin-field QKD network over 127 km of fibre at a secure key rate of 1.398×10^{-5} bits per pulse [31]. Quip Network plans a digital collectible collection generated from verifiable quantum randomness [32].

Identity documents and government credentials have their own thread. NIST published draft materials for adding post-quantum algorithms to federal PIV credentials [33], and GSA is hosting a federal post-quantum summit on 16 September [34]. Giesecke+Devrient is adding prototype post-quantum protocols to its Java Card operating system for European identity cards [35]. GSA's physical access laboratory is expanding to test quantum-resistant building access and employee cards [36].

3. Post-Quantum Migration: Libraries, Platforms and Silicon

Most systems will get post-quantum cryptography from the library or platform they already use, so this is where the migration quietly happens.

The libraries hardened and sped up. Bouncy Castle Java 1.86 closes timing and side-channel leaks across post-quantum algorithms including MAYO, SNOVA and HQC [37]. JDK 27 speeds up Curve25519 field arithmetic, which improves hybrid key exchange throughput by 27 to 51 percent in software [38]. The aws-lc-rs crate made its ML-DSA API stable, including under FIPS [39]. Rustls now enables ML-DSA certificates by default with that provider [40].

wolfSSL added end-to-end SLH-DSA across its stack, including TLS 1.3 and DTLS 1.3 authentication [41]. It also paired with SEALSQ so that wolfTPM can drive the QVault post-quantum TPM in hardware [42]. SEALSQ positions that device as the first shipping TPM 2.0 part with post-quantum algorithms in silicon, which is the vendor's own claim [43]. The Trusted Computing Group is updating its certification programme to cover post-quantum-ready TPMs [44].

The plumbing beneath the libraries was mostly reported on 3 September. wolfPKCS11 2.1 [45] and the OpenSC libp11 provider work [46] were covered then, and no PKCS#11 3.3 revision has appeared [47]. New since then is libp11 0.4.21, which adds provider support for key generation from the OpenSSL command line [46].

On endpoints, Conscrypt 2.7.0 makes the hybrid group the default for TLS key agreement on Android and adds composite ML-DSA and Hash-SLH-DSA signatures [48]. That replaces an April finding that an Android 17 beta lacked post-quantum key exchange entirely [48]. ChromeOS 152 begins moving component and extension delivery signatures from ECDSA to post-quantum algorithms [49]. An out-of-tree Zephyr module brings ML-KEM and ML-DSA to microcontrollers, with benchmarks showing some operations faster than classical baselines [50].

Red Hat's system-wide policies now prefer the hybrid key exchange for libssh applications from RHEL 10.2 [51]. Its documentation warns that the FUTURE policy may one day allow only post-quantum signatures [51]. PostgreSQL 18 exposes TLS group configuration including the hybrid group [52], and Go 1.27.1 ships ML-DSA in the standard library [53]. Both were noted on 3 September.

Firmware roots of trust moved in specification. Caliptra 2.1 adds ML-KEM and an external-mu ML-DSA mode to its Adams Bridge accelerator [54]. The OpenTitan Earl Grey 2 datasheet describes hardened ML-DSA and ML-KEM with masking and fault-injection countermeasures [55]. A September paper, TPMSpy, proposes a platform-agnostic way to trace measured-boot TPM usage on Linux and Windows [56]. SWUpdate now enforces hybrid classical-plus-ML-DSA signing for over-the-air updates, with a build option that blocks downgrades [57].

In silicon, the RISC-V post-quantum extension entered internal review on 11 September, running to 25 September [58]. SEALSQ and the City of Seoul signed a memorandum for a chip security research centre with RISC-V integration in scope [59]. STMicroelectronics' ST54M secure mobile chip completed its EUCC and EMVCo certification testing [60]. IDEMIA topped a Juniper Research post-quantum leaderboard [61].

Unresolved Conflicts

One disagreement sits in this section. The mlkem-native library now ships a native RISC-V vector backend [62]. The ledger's earlier finding was that all RISC-V lattice acceleration remained academic proposals rather than adopted extensions. A software backend on standard vector instructions is not an instruction set extension, so both may be true. The specification review above is what would settle it [58].

4. Post-Quantum Migration: Networks, Cloud and Appliances

Network gear is where post-quantum key exchange first went mainstream, because one appliance can protect every connection behind it.

Cloudflare had the busiest two weeks. Its Automatic Key Exchange, announced 8 September, probes TLS 1.3 origins and prefers the hybrid group on the first ClientHello when the origin supports it [8]. Preferences have been assigned to more than a million domains [8]. Retries fell from about 52 percent to 3.7 percent, and post-quantum origin connections rose to about 45 billion a day [8].

The same measurement found a failure rate. About 0.34 percent of origins fail the handshake when the larger keyshare leads, mostly because middleboxes drop multi-packet ClientHellos [8]. Cloudflare's advice is to audit every device that terminates or inspects TLS on the path to origin [8]. Its resolver at 1.1.1.1 now validates DNSSEC signatures made with ML-DSA-44, at 2,420 bytes each, with full post-quantum DNSSEC planned by 2029 [63]. Cloudflare Radar's public test checks a site for hybrid key exchange support and, since 8 September, for implementation bugs as well [64] [8].

The IETF finished a piece of SSH. RFC 10042, published 31 August, defines the hybrid ML-KEM key exchange methods for SSH [65]. The Internet Architecture Board's October workshop on post-quantum authentication was noted on 3 September [66]. The amortized MLS combiner draft was refreshed [67].

The appliance news reported on 3 September is unchanged. Cisco's IOS XE 26.2 signatures for IKEv2 complete the quantum-safe site-to-site tunnel [68], and Junos widened its SSH protection [69]. F5 validated twelve certificate lifecycle integrations [70], and A10 announced a post-quantum-ready mid-range platform [71]. Arista's EOS 4.36.2F now lists both pure and hybrid ML-KEM groups for control-plane TLS, answering an earlier open question [72]. FortiManager 8.0.0 gains post-quantum key exchange on its management interface through OpenSSL 3.5.5 [73].

RAD's 400 gigabit demarcation device combines ML-KEM hybrid keying with MACsec at line rate, and RAD has partnered with Arqit for key generation [74]. Satellite links joined in. Cyprus and Greece established a quantum-secure satellite connection over Sparkle's interconnect and Hellas Sat [75]. Arqit demonstrated quantum-safe encryption over Es'hailSat's live network [76].

In the cloud, Azure Key Vault Premium and Managed HSM firmware moved to FIPS 140-3 Level 3, correcting an earlier record of 140-2 [77]. Azure Front Door still documents no post-quantum TLS options, which resolves a contradiction on Microsoft's own page in favour of none [78]. NetApp ONTAP 9.17.1 supports post-quantum pre-shared keys for IPsec [79]. Version 9.18.1 negotiates ML-KEM, ML-DSA and SLH-DSA for SSL in both directions when FIPS mode is off [80]. testssl.sh now marks a missing post-quantum KEM as a low-severity finding, a nudge rather than a fault [81].

Unresolved Conflicts

Three disagreements sit in this section. On handshake size, one source says modern stacks negotiate the hybrid group cleanly and the fragmentation problems were early browser rollout issues, now largely fixed [82]. The ledger's Go 1.27 finding says larger ClientHellos still stall middleboxes. Cloudflare's 0.34 percent failure rate suggests both are right about different populations [8].

Netskope's earnings call describes post-quantum algorithms integrated across more than 120 data centres in the second quarter [83]. Its product note describes commercial support behind a tenant-level feature switch [84]. Earlier documentation called the same capability a beta. Release documentation would settle it.

Envoy's BoringSSL backend supports the hybrid group as an opt-in [85]. The ledger's earlier record said Envoy did not support post-quantum cryptography and had closed its tracking issue. A closed issue and an opt-in capability can coexist, and a release note would settle which is current. The Citrix NetScaler question from 3 September, whether default profiles bind the hybrid curve automatically, remains open [86].

5. Post-Quantum Migration: Sectors, Tooling and Governments

Every sector has its own deadlines and its own devices that cannot be patched, and governments are the ones setting the deadlines.

Start with discovery, because every migration starts there. Four CycloneDX releases landed in two weeks. The 2.0 specification milestone incorporated dedicated cryptographic bill of materials features [87], and cyclonedx-go 0.12.0 added the 1.7 cryptographic fields [88]. The editor-validator reached 0.35.0 [89], and a Rust command-line tool now emits a CBOM from a Cargo lockfile [90]. IBM's Quantum Safe Explorer discovers post-quantum algorithms across C#, Java, Python, Go and JavaScript [91] and adds FIPS 140-3 compliance views [92].

Keyfactor passed \$200 million in annual recurring revenue [93]. It also gained ISO 42001 certification for its AI management system [94].

The measurement tools moved as well. SUPERCOP released its 2026.08.31 build and refreshed its result pages [95] [96]. A systematization paper describes TIMECOP as the dynamic tool for catching constant-time

violations, including compiler-induced leaks [97]. A mentorship report on liboqs found a timing leak in NTRU Prime through a nonzero mask function [98].

Sectors next. HHS is preparing migration guidance for hospitals [99]. A healthcare review recommends staged hybrid migration for connected devices, aligned with FDA premarket expectations [100].

In power systems, the IT-heavy portion can migrate by around 2030 while field protocols face a further three to four year delay [101]. A vehicle-to-everything study found Falcon-512 the only NIST signature that fits the sidelink transport block, and only at low traffic [102]. The QORE project completed post-quantum integration for two open 5G cores [103].

Governments spoke. The G7's cybersecurity working group urged immediate migration [9]. The US Treasury set up a Quantum-Readiness Task Force for the financial sector [10], and SEBI's chairman said India's securities markets are preparing [11].

The EU published the results of its consultation on the coordinated roadmap [104], while its Quantum Act slipped to a tentative fourth quarter [105]. ETSI's quantum-safe programme lists new drafts on signatures and PKI and on SLH-DSA implementation [106] [107]. The UK's NCSC pilot runs to March 2027, as reported on 3 September [108].

Unresolved Conflicts

One disagreement sits here. Veridify markets a device-level product supporting post-quantum algorithms for BACnet, Modbus, DNP3 and OPC UA [109]. The ledger's earlier finding was that no vendor-shipped OPC UA integration existed. A vendor page from October 2025 against a later absence finding is a dating question. A product datasheet naming the protocol version would settle it.

6. Quantum Computing: Money, Machines and Theory

Quantum computing news splits into the machines, the money behind them, and the theory that says what they can and cannot do.

The money came first. Four companies turned CHIPS Act possibilities into signed awards. PsiQuantum's is for photonic components [1], Rigetti's for superconducting scaling [2], Quantinuum's for trapped-ion manufacturing with GlobalFoundries [3], and D-Wave's for annealing and gate-model research [4]. PsiQuantum also partnered with Brookhaven National Laboratory on fault-tolerant algorithms [110].

Roadmaps followed. Xanadu targets more than 1,000 logical qubits by 2031, with a qubit factory in 2026 to 2027 and a data centre by 2030 [111]. IonQ's Superion plan has a 256-qubit system delivering in 2027 [112]. Its 10,000-qubit system is to demonstrate fault tolerance in 2027 and sell in 2028 [112].

The public markets are the other funding channel. IonQ raised its 2026 revenue guidance to \$450 million to \$460 million, counting five months of its SkyWater acquisition [113]. At the other end of the market, two pure-play stocks trade under five dollars [114]. For context, two listings are in the pipeline that had not been in the ledger before. SeeQC filed for a traditional IPO in June [115], and Shanghai's TuringQ entered listing tutoring in August, seeking China's first quantum listing [116].

The machines advanced. IBM's Nighthawk r2 adds independent dissipative reset on 120 qubits and runs more than 100,000 circuits a second [6]. It also estimates observables on circuits of more than 7,500 two-qubit gates [6]. IBM separately joined two large cryogenic modules that reached 4 kelvin within five days and cooled below 15 millikelvin [117]. That is a step toward systems larger than one dilution refrigerator [117].

In Jülich, eleQtron's JION trapped-ion machine went live for hybrid use [118], and Maryland researchers proposed trapped-ion gates that need less real-time classical control [119].

Error correction produced three results. Argument reweighting is a postselection strategy for quantum LDPC codes, published in Physical Review Letters [120]. It cuts the logical error rate of the 144-qubit bivariate bicycle code by nearly two orders of magnitude at a rejection rate of only 1.44×10^{-5} [120]. A three-dimensional embedding of the same code on neutral atoms packs logical qubits about four times denser and extracts syndromes about twice as fast [121]. A compiler pass for atom loss cut logical error rates by up to 5.3 times in most tested settings [122].

Theory was busy. One preprint unifies block encoding, qubitization and singular value transformation into a single framework [123]. Another cuts the active volume of Fermi-Hubbard energy estimation by up to 3.9 times [124]. Others prove limits: on one-round quantum local algorithms [125], on query algorithms via Fourier growth [126], and on the semidefinite extension complexity of separable states [127].

A characterisation of one-way puzzles through quantum Kolmogorov complexity [128] and a dequantization result for weighted matrix functions [129] round out the theory. A benchmarking study found that decoder rankings match hardware only when noise models carry per-operation error rates [130]. One applied result adds sixteen features from an eight-qubit circuit to a credit model and lifts its F1 score from 0.462 to 0.517 [131].

7. Post-Quantum Mathematics: Standards and Curves

The mathematics topic follows the algorithms themselves, the standards tracks that ratify them, and the attacks that test them.

HAWK is out. Nine candidates had advanced to the third round of NIST's additional signatures process in May, and HAWK was among them [5]. It was withdrawn on 29 July after a mathematical vulnerability was discovered [5]. Round three continues with the remaining candidates.

ETSI's implementation guidance is filling in. Its TR 104 239 series now has draft parts for ML-KEM, ML-DSA and SLH-DSA at stages from early draft to publication [107]. On the curve side, a 10 September preprint improves supersingularity testing for twisted Hessian curves [132]. It shows that for primes congruent to 2 modulo 3, every ordinary such curve lies on the surface of the 3-isogeny volcano [132].

8. CA/Browser Forum: Root Programs, Ballots and Audits

The Forum and the browser root programs decide what a publicly trusted certificate must look like, and their changes arrive as policy versions and ballots.

The root programs tightened. Apple's Root Program Policy 2.0 took effect on 1 August [133]. It requires new subordinate CAs to carry an extended key usage and to get Apple's approval when externally operated [133].

Chrome's policy 1.8 phases out client authentication: new subordinate CAs must be server-authentication only from 15 June 2026 and new end-entity certificates from 15 March 2027 [134]. It also requires a CA's policy documents to state adherence to the latest published version [135]. Microsoft now requires roots submitted from 1 July 2026 to be dedicated to a single trust purpose [136].

The TLS Baseline Requirements advanced to 2.3.0 through ballot SC100 on DNSSEC clarification, effective 7 September [137]. Redlined and clean versions live in the Forum's repository [138]. Apple's Certificate Transparency policy now accepts logs built on the static-ct-api specification as well as the older RFC [139]. On code signing, Microsoft's Azure Artifact Signing is its recommended non-Store service [140]. Extended validation certificates no longer skip SmartScreen reputation, and Store packages are re-signed automatically [140].

Unresolved Conflicts

Four disagreements sit in this section. Microsoft's root requirements reached version 1.2 on 20 May with stricter key usage controls and mandatory incident reporting [136]. The ledger's earlier record has version 1.1 effective 1 July. A version 1.2 predating 1.1's effective date is possible only if the dates refer to different things, and the document history would settle it.

Ballot NS-010, restructuring the network security requirements, is in its voting period [141]. The earlier record described an August ballot on CA infrastructure inventory and asked for its outcome. Whether these are one ballot or two is unresolved. Mozilla's policy now requires annual mass revocation tests of 30 random certificates within five days, attested from April 2026 [142]. The earlier record placed third-party assessment from June 2025.

The WebTrust applicability matrix names TLS Baseline versions that do not match the published 2.2.9 [143]. A refreshed matrix would settle it.

9. Digital Assets: Ethereum, Bitcoin and the Other Chains

Cryptocurrencies depend on elliptic-curve signatures that a large quantum computer would break, so every chain needs a plan and a date.

Ethereum set its course. The protocol cluster's north star is December 2029 for full quantum resistance across execution, consensus and data layers [144]. Frame transactions, EIP-8141, were named a must-ship proposal for the Hegotá upgrade alongside FOCIL after 62 proposals were reviewed [145]. Frame transactions give accounts programmable validation and a path to post-quantum signatures through key rotation, without a flag day [146]. They moved from considered to scheduled for inclusion on 27 August, and Hegotá moved to 2027 [147].

Research posts explored the state and trie migration implications [148] and payload broadcast rules for larger payloads [149]. Proof systems matter because Ethereum's scaling depends on them. Hash-based STARKs get soundness and zero knowledge from collision resistance alone, so they are plausibly post-quantum [150].

Pairing-based SNARKs lose soundness to Shor's algorithm [150]. The Foundation launched a challenge to raise machine-checked soundness bounds for hash-based proving [151], and no layer-two-specific migration plan has been identified beyond the base-layer roadmap [144].

Bitcoin had its first quantum-safe transaction. A hash-based construction was mined on mainnet on 26 August within existing script limits [152], by StarkWare [153]. Project Eleven's count of exposed coins rose to 8,019,025 BTC, up from 7,039,051 in July [154]. A pre-BIP proposal would standardise how existing outputs are classified by exposure, with a fail-closed rule, to support the migration proposals [155].

Wallets are being measured. Blockstream timed SLH-DSA on today's hardware wallets: key generation in 7 to 16 seconds and signing in about one minute [156], or 53 to 120 seconds by another measurement [157]. Trezor's Safe 7 ships a secure element and bootloader designed for a future post-quantum transition [158]. Fireblocks released an ML-DSA-44 verifier for the EVM at 1.23 million gas, a 6.6-fold cut [159].

Other chains went live. Quantus launched a proof-of-work mainnet with ML-DSA signatures from genesis [160], and QoreChain's Cosmos-based mainnet requires ML-DSA-87 for native transactions [161]. Winternitz one-time signature vaults are live on Solana without a protocol change [162]. Cardano is reviewing an optional post-quantum wallet protection proposal [163].

Institutions moved. The Bitcoin Security Consortium launched with \$15 million pledged over three years by BlackRock, Fidelity Digital Assets, Coinbase, Galaxy, Anchorage and others [164]. The G7 named digital asset handlers in its call to action [9]. The US Treasury's task force names digital assets specifically [10], and SEBI is preparing India's markets [11].

The threat estimates sharpened. One September paper models a trapped-ion machine of about 20,000 physical qubits solving the 256-bit discrete logarithm in about 25.7 days at 63 percent success [165]. That is a single paper's estimate. The ECDSA.Fail challenge cut the space-time cost of the point-addition circuit by 86 percent, to 1,151 logical qubits and about 1.3 million Toffoli gates [166] [167].

Two earlier estimates give those numbers their scale, and both entered the ledger only now. Project Eleven's May report models a baseline Q-Day of 2033, with 2030 optimistic and 2042 pessimistic [168]. A June refinement puts the realistic per-spend attack success nearer 30 percent than the 41 percent best case [169].

Unresolved Conflicts

Two claims about who was first disagree. Algorand executed a fully Falcon-signed mainnet transaction in November 2025 and has recorded more than 140,000 quantum-resistant transactions [170]. The ledger's earlier record held that Algorand's base-layer signing remained Ed25519, with Falcon limited to state proofs.

NEAR activated ML-DSA account signing on mainnet in July [171]. That contradicts the record that Algorand was the only major chain with a NIST-family signature in production. Both may be true if production means different things, and each chain's own documentation of default signing would settle it.

10. Now in the Record

Older material that entered the ledger for the first time in this period, kept apart because none of it is news.

In post-quantum migration, the US classified-solutions programme updated three capability packages in March to add CNSA 2.0 algorithms [172] [173] [174]. Its timeline requires them in all solutions by 2030 [175]. CNSA 2.0 permits LMS and XMSS for firmware signing, with LMS as the default [176]. PCI's HSM standard version 5.0 adds post-quantum definitions and drops TDES [177]. A June Kerberos draft would retire RSA transport and SHA-1 [178], and another adds ML-KEM to PKINIT [179].

The libraries and protocols carry a long tail. OpenSSL 4.0 added an SM2 hybrid group in April [180], and 3.6.3 followed in June [181]. Go 1.27 added a pure ML-KEM-1024 group and ML-DSA constants [182], and Docker builds on Go 1.26 get hybrid exchange by default [183]. RFC 10024 registers the hybrid group as recommended [184], a draft adds ML-KEM-512 hybrids [185], and Alibaba Cloud enables the hybrid group for QUIC by default [186].

libp11 0.4.19 and 0.4.20 added the post-quantum provider operations [46]. The etcd project lags on purpose, adopting Go versions conservatively [187]. mldsa-native 2.0.0 shipped as a formally verified stable release [188], and PQMicroLib-Core gained CAVP certification [189].

Testing and validation also have history. CBOM-Lens was the first producer of CycloneDX 1.7 cryptographic fields [190]. A toolchain automates four constant-time analysers over the signature candidates [191]. Crucible tests 19 implementations in seven languages against real audit findings [192], and NIST enabled the revised ML-KEM and ML-DSA tests in August [193]. Keyfactor's acquisition of Cofide has closed [194].

Platform background is short. wolfSSL 5.9.2 supports Zephyr 4.3 sockets [195]. Akamai offers post-quantum to origin on standard hostnames [196], and Zscaler inspects post-quantum traffic [197]. SAP HANA supports hybrid TLS [198], and Intel is developing ML-DSA attestation for TDX [199]. NetScaler's documentation says the hybrid curve needs an enhanced profile [200].

Sector background follows. The FDA does not yet require a cryptographic bill of materials [100]. Toppan's Java Card supports ML-KEM and ML-DSA, with commercial availability targeted for fiscal 2027 [201]. TCG TPM 2.0 version 1.85 defines the ML-KEM and ML-DSA commands [202].

Galileo's authenticated service started in December 2025, with full encryption planned for late 2026 [203]. Oscilloquartz integrated Galileo authentication into its timing platforms [204]. wolfSSL has a satellite module under test [205], and SKY Perfect JSAT signed a memorandum for a satellite proof of concept [206]. An IEEE framework covers brownfield SCADA migration [207]. Three papers cover surveillance systems [208], cameras [209] and intercoms [210].

GSMA's task force will map every affected specification by year end [211], and a validator framework tests 5G cores [212]. One paper proposes a post-quantum handover for aviation links [213]. A10 spoke at a Taiwan summit [214]. NCSC's July workshop named executive sponsorship and supplier engagement as themes [215].

In finance, SWIFT imposes no signature-size constraint because authentication travels out of band [216]. That retires an earlier recommendation of Falcon-512 for message limits [216]. The same analysis recommends ML-DSA-65 with ML-KEM-768 across payment routes and Falcon-512 only where in-band carriage is required [216]. X9 published a signed QR code standard [217], and EMVCo issued its second quantum position paper [218].

ETSI's signature format standards are moving toward quantum-safe algorithms [219] [220]. An EU regulation fixes the format versions from February 2027 [221].

In quantum computing, Oxford demonstrated fourth-order squeezing in a single trapped ion [222]. Bristol published a fusion-based photonic blueprint [223]. Quantum Motion showed a 99.92 percent readout sensor on an industrial wafer [224]. Three August papers claim advantages: Clifford obfuscation for verifiable circuits [225], a scaling advantage on NP-complete problems [226], and a ten-million-fold measurement reduction using a single sensing qubit [227].

In mathematics, HQC remains pre-draft with no FIPS document as of August [228]. Korea's patent office will run KpqC algorithms from February 2027 [229], and China's industry ministry formed a quantum standards committee [230]. Symmetric cryptanalysis has related-key attacks on Feistel ciphers with complexity reduced by a factor of $2^{129.5}$ [231]. It also has a tight analysis of the Grover-meets-Simon attack [232] and an automated tool for periodic distinguishers [233]. NIST described threshold versions of stateful hash-based signatures [234], an IETF draft adds smaller LMS parameter sets [235], and wolfSSL benchmarked the hash-based families [236].

In the CA/Browser Forum, the EV Guidelines reached 2.0.4 through three ballots [237]. They covered registration numbers [238], domain reuse alignment [239] and clean-up [240]. Microsoft's root requirements version 1.2 and its dedicated-root rule are on file [241], with the GitHub repository as the source of truth [242]. Let's Encrypt will cut lifetimes to 64 days in 2027 and 45 in 2028 [243]. ACME renewal information is RFC 9773 [244] [245].

HARICA revoked 63,525 certificates in days [246], and CRLite reached 87.8 percent coverage [247]. Five RFCs updating certificate handling are recorded [248] [249] [250] [251] [252], with a DNS account challenge draft [253] and a CAA parameter requirement from March 2027 [254]. On audits, the record holds WebTrust network security 2.0.5 [255], ETSI EN 319 401 [256] and 411-1 [257], and the ACAB's letter templates [258]. It also holds Microsoft's attestation letter rules [259] and the three-month audit publication rule [260]. The Forum adopted IPR Policy 1.4 [261], effective 1 June [262].

In digital assets, STARKs are the practical post-quantum path for credentials [263]. The CAPSS framework turns hash permutations into signatures of 9.5 to 15.5 kilobytes [264]. QRL's second testnet runs ML-DSA-87 through the full stack [265], and Nervos CKB has a SPHINCS+ lock script in production [266]. XRPL works with Project Eleven on audits and custody [267], and Cardano's research programme includes a post-quantum initiative [268]. Polkadot's roadmap proposes ML-DSA and Falcon [269], and Circle's Arc chain plans quantum-secure signatures from launch [270] [271].

A hardware wallet signs every transaction with SLH-DSA [272]. Presidio published a living readiness paper [273]. An educational BIP-360 implementation passes all vectors [274], and Blockstream's SHRINCS candidates give 5.7 kilobyte fallback signatures [275].

Four papers cover post-quantum HD wallets: a lattice construction with full derivation [276] and an ML-DSA one with hardened derivation only [277]. The other two are zero-knowledge schemes for recovering unmigrated funds [278] [279]. Cyber insurers are discussing migration coverage [280].

About This Report

The Spaced Research Ledger (SRL) is an independent, not-for-profit research project. It is an automated system that gathers claims from live public sources, checks each one against its origin, and records every disagreement instead of merging it. Every stage runs on a different AI model, drawn from three to seven systems across competing labs, so no single model both finds a claim and judges it. The method is subject-neutral: the same pipeline can be pointed at any narrow field, gather what public sources currently say about it, and keep that picture current as the field moves.

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